Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	*(a)*	(i)	Velocity [of car] is changing [with time] (1) because its direction is changing [so the car is accelerating] (1) **or** There is a <u>resultant</u> force (1) [towards the centre] due to friction / grip (1)	2			2		
		(ii)	$v = \frac{45\times10^3}{60\times60}$ = 12.5 m s^{-1} (conversion) (1) $\omega = \frac{v}{r} = \frac{12.5}{80}$ = 0.156 rad s^{-1} substitution and calculation (1)		2		2	2	
		(iii)	$a = \frac{v^2}{r} = \frac{(12.5)^2}{80}$ substitution (**ecf**) (1) [Alt: use $a=\omega^2 r$] = 1.95 m s^{-2} (1) Direction: towards centre (of 'circular' motion) (1)	1 1	1		3	2	
	(b)		Either: Any two × (1) of these points - Appropriate tyre design for friction - Banking of road [for contribution from normal contact force] - Appropriate surface - Suspension set-up - Anti-roll bars. (or any sensible answers, one referring to road and the other to the car) **Or** any one sensible point (1) + explanation of the role of physics (1) [			2	2		
			Question 1 total	**4**	**3**	**2**	**9**	**4**	**0**